5 **(a)** Destructive interference is when <u>two waves arrive at the same point anti-phase</u> (phase difference of π / 3π etc rad) they superpose to produce a <u>resultant wave with minimum (or zero) amplitude or a minimum is produced.</u> B1 B1

(b) **(i)**

$$\lambda = \frac{v}{f}$$

$$\lambda = \frac{v}{f} = \frac{330}{1780} = 0.185m$$ A1

(ii)

$$S_1D = \sqrt{12^2 + 4^2} = 12.649 \text{ m}$$

Path difference

= 12.649 – 12

$$= \frac{0.649}{0.18539}\lambda$$ C1

$$= 3.5\lambda$$ A1

(iii) The path difference of 3.5λ would lead to a phase difference of π rad. Since sound from both speakers have a phase difference of π rad, this would mean that there is no net phase difference at D. B1

Constructive interference of sound waves will take place and loud sound will be heard at D. B1

(c) **(i)** d sinθ = nλ

Since sin θ cannot be greater than 1

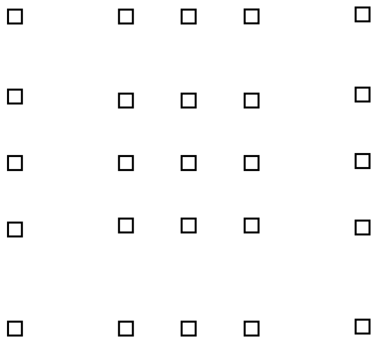
For max order, sin θ < 1

n λ / d <1

n < 2.7 C1

Since n must be an integer, and cannot exceed 2.7, the max order =**<u>2</u>** A1

(ii)



1 mark for correct shape grid of spots 5x5 B1
1 mark for 2nd order dots are further apart than 1st order dots B1

(iii) The new light source has shorter wavelength. Using $d \sin\theta = n\lambda$, the diffracted angle for each wave will be smaller.

Hence, more bright spots can be seen on the screen and the closer spacing between spots. B1

6 (a) (i) $100\pi = 2\pi f$, so frequency $f = 50$ Hz B1